ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2021 P2H Worked Solutions

Jan 2021 | P2H

25 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2021 P2H - Jan 2021 | P2H

1 A train takes 6 hours 39 minutes to travel from New Delhi to Kanpur.
The train travels a distance of 429 km.

Work out the average speed of the train.
Give your answer in km/h correct to one decimal place.

....km/h

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION Q#: 1 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. The time is

The time is

$$6 \text{ h } 39 \text{ min} = 6.65 \text{ h}$$

2. Average speed

Average speed:

$$429 \div 6.65 = 64.511\ldots$$

FINAL ANSWER

64.5 km/h.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

2 Ava writes down five whole numbers.

For these five numbers

the median is 7

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Calculate statistic

The median is 7, so the middle number is 7.

2. Calculate statistic

The mode is 8, so use two 8s.

3. Calculate statistic

The range is 5, so choose the smallest and largest numbers to differ by 5.
One possible set is

$$3,\ 6,\ 7,\ 8,\ 8$$

4. Range

Range:

$$8 - 3 = 5$$

5. Mode

Mode:

$$8$$

6. Median

Median:

$$7$$

FINAL ANSWER
$3,\ 6,\ 7,\ 8,\ 8.$

PAST PAPER QUESTION

Q#: 3 Marks: 6

TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

3 Gladys buys a table for $465 to sell in her shop.

She sells the table for $520

(a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

.... %

(3)

Gladys has a sale in her shop.

She decreases all the normal prices by 12%
The normal price of an armchair was $550

(b) Work out the sale price of the armchair.

$ ....

(3)

(Total for Question 3 is 6 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 6

TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Profit

Profit:

$$520 - 465 = 55$$

2. Percentage profit

Percentage profit:

$$\frac{55}{465} \times 100 = 11.827\ldots$$

3. Correct to 3 significant figures

Correct to 3 significant figures:

$$11.8\%$$

4. Calculate statistic

For the sale price, a 12% decrease means 88% of the original price:

$$550 \times 0.88 = 484$$

FINAL ANSWER

(a) 11.8%, (b) $484

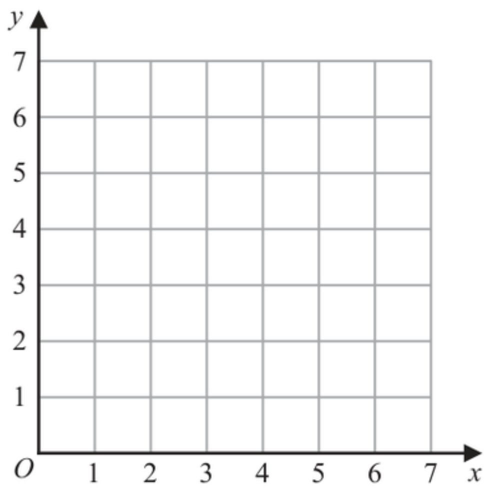
PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Graphing Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$x \geqslant 1.5$ $\quad$ $y \geqslant x$ $\quad$ $x + y \leqslant 6$

Label the region **R**.

(1)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Graphing Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Draw the three boundary lines

Draw the three boundary lines:

$$x = 1.5, \qquad y = x, \qquad x + y = 6$$

2. The required region satisfies

The required region satisfies

$$x > 1.5$$

$$y \geq x$$

$$x + y \leq 6$$

FINAL ANSWER

Shade the region $x > 1.5$, $y \geq x$, $x + y \leq 6$.

PAST PAPER QUESTION

Q#: 5 Marks: 7

TOPIC: **Solving Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

5 (a) Expand and simplify $4x(2x+5) - 3x(2x-3)$

....

(2)

Given that $\dfrac{y^5 \times y^n}{y^6} = y^{13}$

(b) work out the value of n.

$n =$ ....

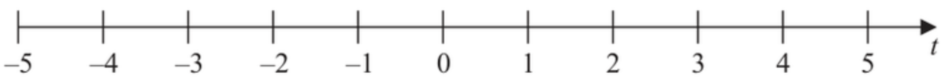
(2)

(c) (i) Solve the inequality $7t - 8 < 2t + 7$

....

(2)

(ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

(Total for Question 5 is 7 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 7

TOPIC: **Solving Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. (a)

(a)

$$4x(2x+5)-3x(2x-3)=8x^2+20x-6x^2+9x=2x^2+29x$$

2. Expand brackets

(b)

$$\frac{y^5\times y^n}{y^6}=y^{n-1}=y^{13}$$

so $n-1=13$, hence $n=14$.

3. (c)

(c)

$$7t-8<2t+7$$

$$5t<15$$

$$t<3$$

4. Expand brackets

On the number line, use an open circle at 3 and shade to the left.

FINAL ANSWER

(a) $2x^2+29x$, (b) $n=14$, (c) $t<3$

PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jan 2021 P2H - Jan 2021 | P2H

6 (a) Write down the value of y^0

....
(1)

(b) Work out $\dfrac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

....
(3)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Convert standard form

Any non-zero value to the power 0 is 1, so

$$y^0 = 1$$

2. Convert standard form

For part (b),

$$9.6 \times 10^{141} + 6.4 \times 10^{140} = 9.6 \times 10^{141} + 0.64 \times 10^{141}$$

$$= 10.24 \times 10^{141} = 1.024 \times 10^{142}$$

3. Convert standard form

Now divide:

$$\frac{1.024 \times 10^{142}}{3.2 \times 10^{16}} = 0.32 \times 10^{126}$$

$$= 3.2 \times 10^{125}$$

FINAL ANSWER

(a) 1, (b) 3.2×10^{125}

PAST PAPER QUESTION Q#: 7 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

7 There are 5 cocoa pods in a bag.
The mean weight of the 5 cocoa pods is 398 grams.

A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.

Work out the weight of the sixth cocoa pod that is put into the bag.

.... grams

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION Q#: 7 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Total weight of 5 cocoa

Total weight of 5 cocoa pods:

$$5 \times 398 = 1990 \text{ g}$$

2. Total weight of 6 cocoa

Total weight of 6 cocoa pods:

$$6 \times 401 = 2406 \text{ g}$$

3. Weight of the sixth cocoa

Weight of the sixth cocoa pod:

$$2406 - 1990 = 416$$

FINAL ANSWER

416 grams.

PAST PAPER QUESTION

Q#: 8 Marks: 5

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

8 *A*, *B* and *C* are points on a circle with centre *O*.

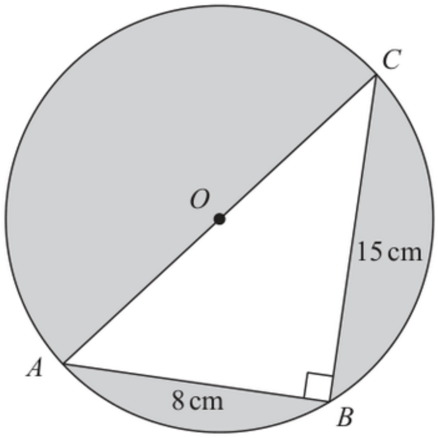

Diagram **NOT** accurately drawn

AOC is a diameter of the circle.

$AB = 8$ cm $\quad BC = 15$ cm

Angle $ABC = 90°$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION Q#: 8 Marks: 5

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Triangle is right angled

Triangle ABC is right-angled:

$$AC^2 = 8^2 + 15^2 = 64 + 225 = 289$$

$$AC = 17 \text{ cm}$$

2. Calculate area

Since AOC is a diameter, the radius of the circle is

$$17 \div 2 = 8.5 \text{ cm}$$

3. Area of the circle

Area of the circle:

$$\pi(8.5)^2 = 226.98\ldots \text{ cm}^2$$

4. Area of triangle

Area of triangle ABC:

$$\frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

5. The shaded area is

The shaded area is

$$226.98\ldots - 60 = 166.98\ldots$$

FINAL ANSWER

167 cm^2.

PAST PAPER QUESTION

Q#: 9 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jan 2021 P2H - Jan 2021 | P2H

9

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$
$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of powers of prime numbers.

(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION Q#: 9 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Find the LCM

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

2. Find the LCM

For the LCM, take the highest power of every prime that appears:

$$\text{LCM} = 2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

FINAL ANSWER

$2^4 \times 3^2 \times 5^4 \times 11 \times 13$

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

10 The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION Q#: 10 Marks: 3

TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. The ratio of Team A

The ratio of Team A to Team B is 3 : 4.

2. Use parts for Team A

Use 3 parts for Team A and 4 parts for Team B.

3. Full time part from Team

Full-time part from Team A:

$$\frac{4}{5} \times 3 = \frac{12}{5}$$

4. Full time part from Team

Full-time part from Team B:

$$24\% \times 4 = \frac{24}{100} \times 4 = \frac{24}{25}$$

5. Total full time parts

Total full-time parts:

$$\frac{12}{5} + \frac{24}{25} = \frac{60}{25} + \frac{24}{25} = \frac{84}{25}$$

6. Total parts

Total parts:

$$3 + 4 = 7$$

7. Simplify fraction

So the fraction who work full time is

$$\frac{\frac{84}{25}}{7} = \frac{84}{175} = \frac{12}{25}$$

FINAL ANSWER

$$\frac{12}{25}$$

PAST PAPER QUESTION

Q#: 11 **Marks: 3**

TOPIC: **Algebraic Roots & Indices**

Jan 2021 P2H - Jan 2021 | P2H

11 Simplify fully $\left(\dfrac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Use index laws

$$\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$$

2. Simplify inside the brackets

Simplify inside the brackets:

$$\frac{9t^4w^9}{18t^6w^{10}} = \frac{1}{2}t^{-2}w^{-1} = \frac{1}{2t^2w}$$

3. Use index laws

Now apply the power -2:

$$\left(\frac{1}{2t^2w}\right)^{-2} = (2t^2w)^2$$

$$= 4t^4w^2$$

FINAL ANSWER

$4t^4w^2$.

PAST PAPER QUESTION

Q#: 12 Marks: 2

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

12 15 people were asked how long, in minutes, they had been waiting for a bus.

Here are the results.

2	3	3	4	5	6	6	8	9	10	11	13	14	15	18

Find the interquartile range of these times.

.... minutes

(Total for Question 12 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 12 Marks: 2

TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Use Statistics Toolkit correct

The data is already in order:

$$2, 3, 3, 4, 5, 6, 6, 8, 9, 10, 11, 13, 14, 15, 18$$

2. Calculate value

For 15 values,

$$Q_1 = 4, \qquad Q_3 = 13$$

$$\text{IQR} = 13 - 4 = 9$$

FINAL ANSWER

9 minutes.

PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Circle Theorems**

Jan 2021 P2H - Jan 2021 | P2H

13 P, Q, R, S and T are points on a circle with centre O.

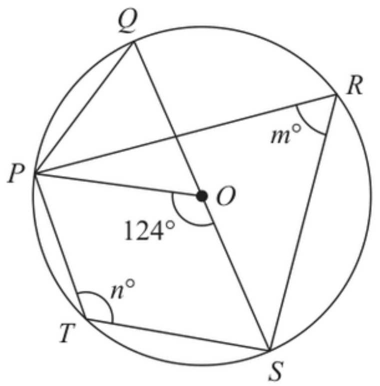

Diagram **NOT** accurately drawn

QOS is a diameter of the circle.

angle $POS = 124°$ angle $PRS = m°$ angle $PTS = n°$

(a) Find the value of

(i) m

....

(ii) n

....

(2)

(b) Find the size of angle QPO.

.... °

(1)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Circle Theorems**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. Use trigonometry

The central angle $\angle POS$ is 124°. The angle at the circumference standing on the same chord PS is half this angle.

$$m = \frac{124}{2} = 62$$

2. Use circle theorem

Angles opposite each other in a cyclic quadrilateral add to 180°, so

$$n = 180 - 62 = 118$$

3. Use trigonometry

Since QOS is a diameter,

$$\angle QOP = 180^\circ - 124^\circ = 56^\circ$$

4. Use trigonometry

Triangle QOP is isosceles because $OQ = OP$. Therefore

$$\angle QPO = \frac{180 - 56}{2} = 62^\circ$$

FINAL ANSWER

$m = 62$, $n = 118$, and $\angle QPO = 62^\circ$.

PAST PAPER QUESTION Q#: 14 Marks: 7

TOPIC: **Rearranging Formulas**

Jan 2021 P2H - Jan 2021 | P2H

14 (a) Solve $\dfrac{9a-7}{5} - \dfrac{3a-7}{4} = 4.55$

Show clear algebraic working.

$a =$ ....

(3)

(b) Make c the subject of the formula $p = \sqrt{\dfrac{ac+8}{3+c}}$

....

(4)

(Total for Question 14 is 7 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 7

TOPIC: **Rearranging Formulas**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a)

$$\frac{9a-7}{5}-\frac{3a-7}{4}=4.55$$

2. Multiply by 20

Multiply by 20:

$$4(9a-7)-5(3a-7)=91$$

$$36a-28-15a+35=91$$

$$21a+7=91$$

$$a=4$$

3. (b)

(b)

$$p=\sqrt{\frac{ac+8}{3+c}}$$

$$p^2=\frac{ac+8}{3+c}$$

$$p^2(3+c)=ac+8$$

$$c(p^2-a)=8-3p^2$$

FINAL ANSWER

(a) $a=4$, (b) $c=\frac{8-3p^2}{p^2-a}$

PAST PAPER QUESTION

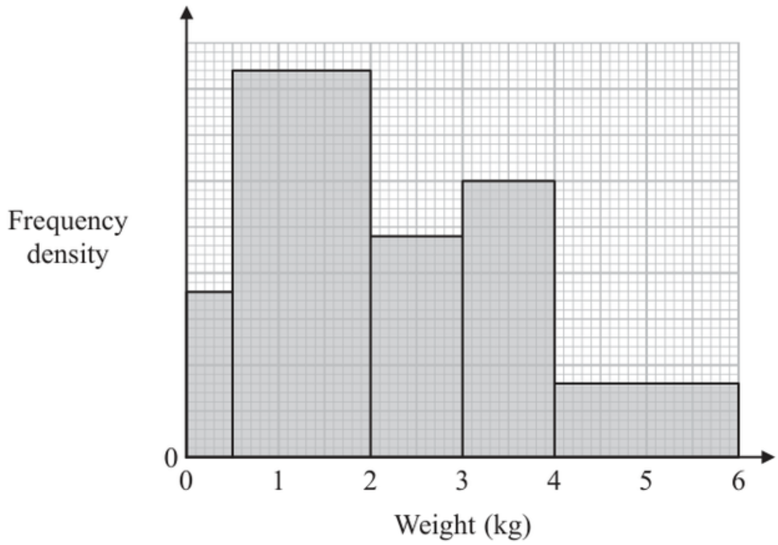
Q#: 15 Marks: 6

TOPIC: **Histograms**

Jan 2021 P2H - Jan 2021 | P2H

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

....

(3)

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

....

(3)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 6

TOPIC: **Histograms**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use histogram

For $0.5 < w \leq 2$, the frequency is 63, so the density is

$$\frac{63}{1.5} = 42$$

2. Use histogram

Using the histogram heights, the class frequencies are

$$9,\ 63,\ 24,\ 30,\ 16$$

3. Use histogram

So the total number of parcels is

$$9 + 63 + 24 + 30 + 16 = 142$$

4. Use histogram

For weight greater than 2.25 kg:

$$0.75(24) + 30 + 16 = 64$$

5. The probability that both randomly

The probability that both randomly chosen records are greater than 2.25 kg is

$$\frac{64}{142} \times \frac{63}{141} = 0.201\ldots$$

FINAL ANSWER

142, then 0.201 approximately.

PAST PAPER QUESTION

Q#: 16 Marks: 6

TOPIC: **Set Notation & Venn Diagrams**

Jan 2021 P2H - Jan 2021 | P2H

16 Some students were asked the following question.

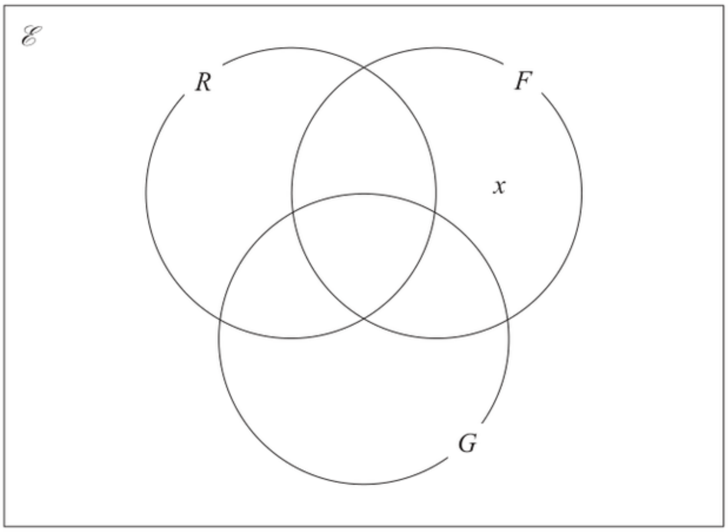
"Which of the subjects Russian (R), French (F) and German (G) do you study?"

Of these students

4 study all three of Russian, French and German
10 study Russian and French
13 study French and German
6 study Russian and German
24 study German
11 study none of the three subjects
the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

....

(3)

(Total for Question 16 is 6 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 6

TOPIC: **Set Notation & Venn Diagrams**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. The number studying all three

The number studying all three languages is 4.

2. Calculate value

So the two-language-only regions are

$$R \cap F \text{ only} = 10 - 4 = 6$$

$$F \cap G \text{ only} = 13 - 4 = 9$$

$$R \cap G \text{ only} = 6 - 4 = 2$$

3. The German only region is

The German-only region is

$$24 - 9 - 2 - 4 = 9$$

4. The French only region is

The French-only region is x and the Russian-only region is $2x$. There are 11 students outside all three sets.

5. Calculate value

Using the total 80,

$$2x + x + 9 + 6 + 9 + 2 + 4 + 11 = 80$$

$$3x + 41 = 80$$

$$x = 13$$

6. The number studying Russian is

The number studying Russian is

$$2x + 6 + 2 + 4 = 26 + 6 + 2 + 4 = 38$$

FINAL ANSWER

38

PAST PAPER QUESTION

Q#: 17 Marks: 7

TOPIC: **Volume & Surface Area**

Jan 2021 P2H - Jan 2021 | P2H

17 The diagram shows a solid prism *ABCDEFGH*.

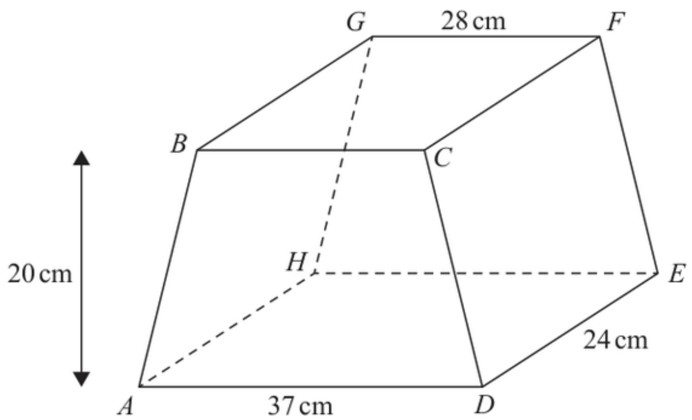

Diagram **NOT** accurately drawn

The trapezium *ABCD*, in which *AD* is parallel to *BC*, is a cross section of the prism.
The base *ADEH* of the prism is a horizontal plane.
ADEH and *BCFG* are rectangles.
The midpoint of *BC* is vertically above the midpoint of *AD* so that $BA = CD$.

$AD = 37$ cm $\quad$ $GF = 28$ cm $\quad$ $DE = 24$ cm

The perpendicular distance between edges *AD* and *BC* is 20 cm.

(a) Work out the total surface area of the prism.

.... cm²

(4)

(b) Calculate the size of the angle between *AF* and the plane *ADEH*.
Give your answer correct to one decimal place.

.... °

(3)

(Total for Question 17 is 7 marks)

PAST PAPER SOLUTION Q#: 17 Marks: 7

TOPIC: **Volume & Surface Area**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

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1. The cross section is a

The cross-section is a trapezium with parallel sides 37 cm and 28 cm, and perpendicular height 20 cm.

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

2. Calculate area

Since the trapezium is isosceles, the horizontal difference on each side is

$$\frac{37 - 28}{2} = 4.5$$

3. Calculate area

So

$$AB = CD = \sqrt{20^2 + 4.5^2} = 20.5$$

4. The perimeter of the cross

The perimeter of the cross-section is

$$37 + 28 + 20.5 + 20.5 = 106$$

5. The length of the prism

The length of the prism is 24 cm, so the total surface area is

$$2(650) + 106(24) = 3844$$

6. Use trigonometry

For the angle between AF and the horizontal plane, the horizontal projection has length

$$\sqrt{24^2 + 4.5^2}$$

7. Use trigonometry

The vertical height is 20, so

$$\tan\theta = \frac{20}{\sqrt{24^2 + 4.5^2}}$$

$$\theta = 39.3^\circ$$

FINAL ANSWER

3844 cm^2, and 39.3°.

PAST PAPER QUESTION

Q#: 18 Marks: 7

TOPIC: **Coordinate Geometry**

Jan 2021 P2H - Jan 2021 | P2H

18 A rectangle *ABCD* is to be drawn on a centimetre grid such that

A has coordinates (–4, –2)
B has coordinates (1, 10)
C has coordinates (19, *a*)
D has coordinates (*b*, *c*)

(a) Work out the value of *a*, the value of *b* and the value of *c*.

$a =$ ....

$b =$ ....

$c =$ ....

(4)

(b) Calculate the perimeter, in centimetres, of rectangle *ABCD*.

.... cm

(3)

(Total for Question 18 is 7 marks)

PAST PAPER SOLUTION Q#: 18 Marks: 7

TOPIC: **Coordinate Geometry**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Calculate value

$$A = (-4, -2), \quad B = (1, 10), \quad C = (19, a), \quad D = (b, c)$$

2. The vector is

The vector AB is

$$(1 - (-4), 10 - (-2)) = (5, 12)$$

3. The vector is

The vector BC is

$$(19 - 1, a - 10) = (18, a - 10)$$

4. Adjacent sides of a rectangle

Adjacent sides of a rectangle are perpendicular, so their dot product is zero:

$$5(18) + 12(a - 10) = 0$$

$$90 + 12a - 120 = 0$$

$$a = \frac{5}{2}$$

5. Calculate value

Now

$$BC = (18, -7.5)$$

6. Calculate value

So

$$D = A + BC = (-4, -2) + (18, -7.5) = (14, -9.5)$$

$$b = 14, \qquad c = -9.5$$

7. The side lengths are

The side lengths are

$$AB = \sqrt{5^2 + 12^2} = 13$$

$$BC = \sqrt{18^2 + (-7.5)^2} = 19.5$$

8. Perimeter

Perimeter:

$$2(13 + 19.5) = 65$$

FINAL ANSWER

$a = \frac{5}{2}$, $b = 14$, $c = -9.5$, and perimeter 65 cm.

PAST PAPER QUESTION

Q#: 19 Marks: 5

TOPIC: **Differentiation**

Jan 2021 P2H - Jan 2021 | P2H

19 A particle P is moving along a straight line.
The fixed point O lies on this line.

At time t seconds where $t \geqslant 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

.... metres

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 **Marks: 5**

TOPIC: **Differentiation**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate statistic

$$s = t^3 + 5t^2 - 8t + 10$$

2. Velocity is

Velocity is

$$\frac{ds}{dt} = 3t^2 + 10t - 8$$

3. Calculate statistic

At rest,

$$3t^2 + 10t - 8 = 0$$

4. Factorise

Factorise:

$$(3t - 2)(t + 4) = 0$$

5. Calculate statistic

So

$$t = \frac{2}{3} \quad \text{or} \quad t = -4$$

6. Calculate statistic

Since $t \geq 0$,

$$t = \frac{2}{3}$$

7. Calculate statistic

Now find s:

$$s = \left(\frac{2}{3}\right)^3 + 5\left(\frac{2}{3}\right)^2 - 8\left(\frac{2}{3}\right) + 10$$

$$s = \frac{8}{27} + \frac{20}{9} - \frac{16}{3} + 10$$

$$s = \frac{194}{27}$$

FINAL ANSWER
$\frac{194}{27}$ metres.

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

20

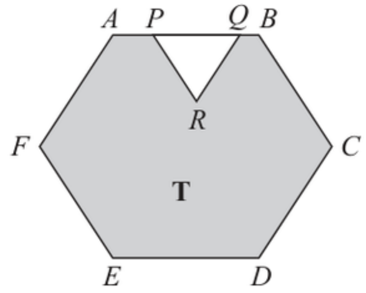

Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm²

work out the length of PQ.

.... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let

Let

$$PQ = p$$

2. Given

Given

$$AB = 1.5PQ$$

3. Find the gradient

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

4. Area of a regular hexagon

Area of a regular hexagon with side length s is

$$\frac{3\sqrt{3}}{2}s^2$$

5. Find the gradient

So the area of the hexagon is

$$\frac{3\sqrt{3}}{2}\left(\frac{3}{2}p\right)^2$$

$$= \frac{27\sqrt{3}}{8}p^2$$

6. Find the gradient

The removed triangle PQR is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

7. The shaded area is

The shaded area is

$$\frac{27\sqrt{3}}{8}p^2 - \frac{\sqrt{3}}{4}p^2$$

$$= \frac{25\sqrt{3}}{8}p^2$$

8. Find the gradient

This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

FINAL ANSWER

$PQ = 4.8$ cm

PAST PAPER QUESTION **Q#: 20** **Marks: 4**

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

20

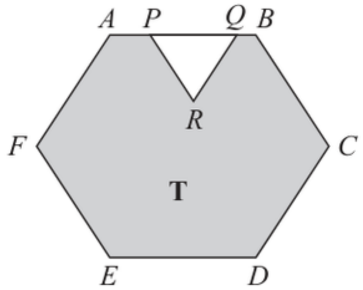

Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm^2

work out the length of PQ.

.... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let

Let

$$PQ = p$$

2. Given

Given

$$AB = 1.5PQ$$

3. Find the gradient

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

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6. Find the gradient

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$$= \frac{25\sqrt{3}}{8}p^2$$

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This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

FINAL ANSWER

$PQ = 4.8$ cm

PAST PAPER QUESTION

Topic: **Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

Q#: 21 Marks: 4

21 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

$$\frac{25x^2-64}{5x^2-13x-6}\times\frac{x^2-8x+15}{5x+8}-(x-7)$$

2. Factor

Factor:

$$25x^2-64=(5x-8)(5x+8)$$

$$5x^2-13x-6=(5x+2)(x-3)$$

$$x^2-8x+15=(x-3)(x-5)$$

3. Use trigonometry

So

$$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)}\times\frac{(x-3)(x-5)}{5x+8}-(x-7)$$

4. Cancel common factors

Cancel common factors:

$$\frac{(5x-8)(x-5)}{5x+2}-(x-7)$$

5. Use the common denominator

Use the common denominator $5x+2$:

$$\frac{(5x-8)(x-5)-(x-7)(5x+2)}{5x+2}$$

6. Expand

Expand:

$$(5x-8)(x-5)=5x^2-33x+40$$

$$(x-7)(5x+2)=5x^2-33x-14$$

7. Use trigonometry

So the numerator is

$$5x^2 - 33x + 40 - (5x^2 - 33x - 14) = 54$$

FINAL ANSWER

$$\boxed{\frac{54}{5x+2}}$$

PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

21 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 4

TOPIC: **Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

$$\frac{25x^2-64}{5x^2-13x-6} \times \frac{x^2-8x+15}{5x+8} - (x-7)$$

2. Factor

Factor:

$$25x^2 - 64 = (5x-8)(5x+8)$$

$$5x^2 - 13x - 6 = (5x+2)(x-3)$$

$$x^2 - 8x + 15 = (x-3)(x-5)$$

3. Use trigonometry

So

$$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)} \times \frac{(x-3)(x-5)}{5x+8} - (x-7)$$

4. Cancel common factors

Cancel common factors:

$$\frac{(5x-8)(x-5)}{5x+2} - (x-7)$$

5. Use the common denominator

Use the common denominator $5x+2$:

$$\frac{(5x-8)(x-5) - (x-7)(5x+2)}{5x+2}$$

6. Expand

Expand:

$$(5x-8)(x-5) = 5x^2 - 33x + 40$$

$$(x-7)(5x+2) = 5x^2 - 33x - 14$$

7. Use trigonometry

So the numerator is

$$5x^2 - 33x + 40 - (5x^2 - 33x - 14) = 54$$

FINAL ANSWER

$$\boxed{\frac{54}{5x+2}}$$

PAST PAPER QUESTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

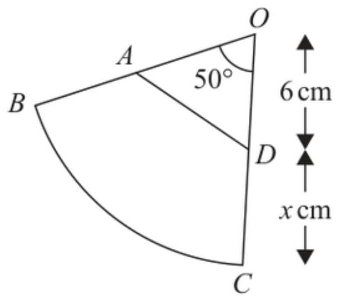

Diagram **NOT** accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x.
Give your answer correct to 3 significant figures.

$x =$ ....

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The radius of sector is

The radius of sector OBC is

$$6+x$$

2. Use trigonometry

The angle is $50°$, so the arc length of sector OBC is

$$\frac{50}{360}\times 2\pi(6+x)$$

3. Evaluate fraction

Therefore the perimeter of sector OBC is

$$2(6+x)+\frac{50}{360}\times 2\pi(6+x)$$

$$=(6+x)\left(2+\frac{5\pi}{18}\right)$$

4. Use trigonometry

In triangle OAD,

$$OA=OD=6,\qquad \angle AOD=50^\circ$$

5. Use trigonometry

So

$$AD=2(6)\sin 25^\circ=12\sin 25^\circ$$

6. The perimeter of triangle is

The perimeter of triangle OAD is

$$12+12\sin 25^\circ$$

7. Given that

Given that

$$\text{perimeter of sector } OBC = 2\times\text{perimeter of triangle } OAD$$

$$(6+x)\left(2+\frac{5\pi}{18}\right)=2(12+12\sin 25^\circ)$$

$$x = \frac{2(12 + 12\sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854\ldots$$

FINAL ANSWER

$\boxed{5.89}$

PAST PAPER QUESTION Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

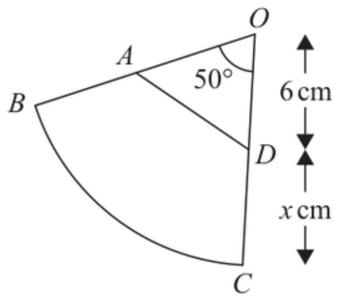

Diagram **NOT** accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x.
Give your answer correct to 3 significant figures.

$x =$ ....

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The radius of sector is

The radius of sector OBC is

$$6 + x$$

2. Use trigonometry

The angle is $50°$, so the arc length of sector OBC is

$$\frac{50}{360} \times 2\pi(6 + x)$$

3. Evaluate fraction

Therefore the perimeter of sector OBC is

$$2(6 + x) + \frac{50}{360} \times 2\pi(6 + x)$$
$$= (6 + x)\left(2 + \frac{5\pi}{18}\right)$$

4. Use trigonometry

In triangle OAD,

$$OA = OD = 6, \qquad \angle AOD = 50°$$

5. Use trigonometry

So

$$AD = 2(6)\sin 25° = 12\sin 25°$$

6. The perimeter of triangle is

The perimeter of triangle OAD is

$$12 + 12\sin 25°$$

7. Given that

Given that

$$\text{perimeter of sector } OBC = 2 \times \text{perimeter of triangle } OAD$$

$$(6 + x)\left(2 + \frac{5\pi}{18}\right) = 2(12 + 12\sin 25°)$$

$$x = \frac{2(12 + 12\sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854\ldots$$

FINAL ANSWER

$\boxed{5.89}$